CO1 - Assessment

Name : B. Akshaya
Reg no : 192424336

Coursecode : DSA0607
Coursename : Data Handling & Visualization

Test

1) Concept of Mapping data

Introduction :

Representing the data by graphs to under the data clearly and exactly what data is showing.

Aesthetics in data visualization are :

* Here Mapping data on a asethetics is the process of assigning data values to usual the properties such as

* position
* size
* shape
* colour

Here, these four aesthetics shows the meaningful understand for the data.

For Example :

In a class students, girls, and boys of their marks shown.

position : shows where data points are placed

- x-axis - students
- y-axis - Marks

Size : Represents the magnitude of values

Shape : Differentiative categories

□ - girls
Δ - boys

Colour : Represents the Categories of numerical values.

red - high marks
blue - low marks

There are different types of representing data :

1) categorical data : consistency of numerical values that represents the groups of attributes

2) Ordinal data : consistency of data which contain the ordinary information.

3) Nominal data

4) Quantative data

1) categorical data :

Example : gender based

blood group based

2) ordinal data :

Example : low medium high

2) "Role of scale in data visualization" explain the how scale role in data visualization.

- scale is the keyrole in data vizualization
- scale represents the categories as the
  - x-axis
  - y-axis

* A scale converts the "data into visuals elements" to understand.

there are aesthetics are :

position, color, size and shape

1. linear scale: It represents the Equal spacing values

Example :

50, 100, 150, 200 etc. . . .

2. Logorthmic scale : It is used for very large ranges data as log

Example :

10 , 100, 1000, 10000 etc...

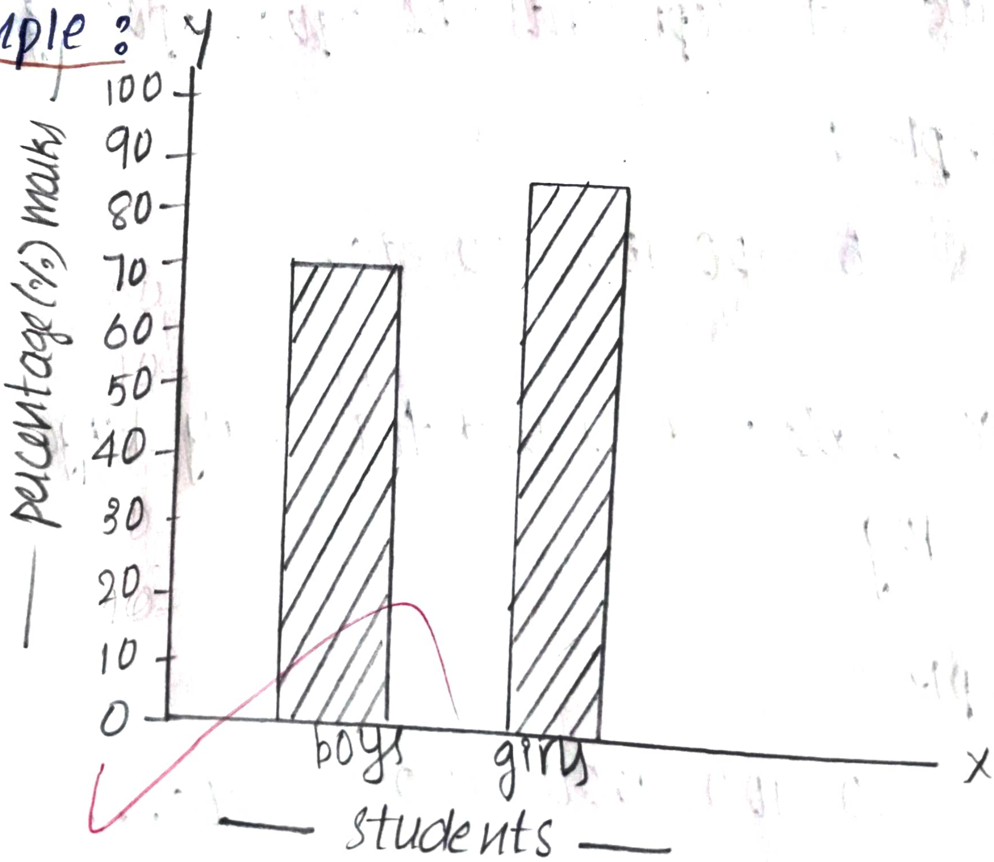
3. 3. colour scale : It is used for scaling data with different colours to understand the data

Example:
red, blue etc...

Importance:

* Ensure accurate representation
* Make graphs easy to read
* Helps to understand clearly
* Helps for data values to represent as element

Example:



Conclusion:

choosing the correct scale is essential for accurate and effective visualization.

"Role of color in data visualization"

- colour is the asthetic in the "data handling"
- colour is the one of the important asethetic in the all of four asethetics in the data visualization.
- "colour represents the distinguish categories" and represent the data values and the data highlights
- colour represents used in the graph as heat ma for temperature. It shows:
  * red - as highly temperature
  * thick blue - as normal temperature
  * blue - low temperature

1) distinguish colour:

- It is used for the distinguish categories as the different groups.

Example: HR (Blue colour)

2) Representing data values

- It shows for the colour gradients show increasi or decreasing

Example :

Blue → cold
Red → Hot

principles of Effective colours are:

- contrast : measure readability
- Accessability sure for colorblind
- visual harmony : used for balanced & consistent

- If colours are used more in graphs it shows the confuse in the graph as data representing.

So, use only light colors and thick colors together